

Ex.no:7a

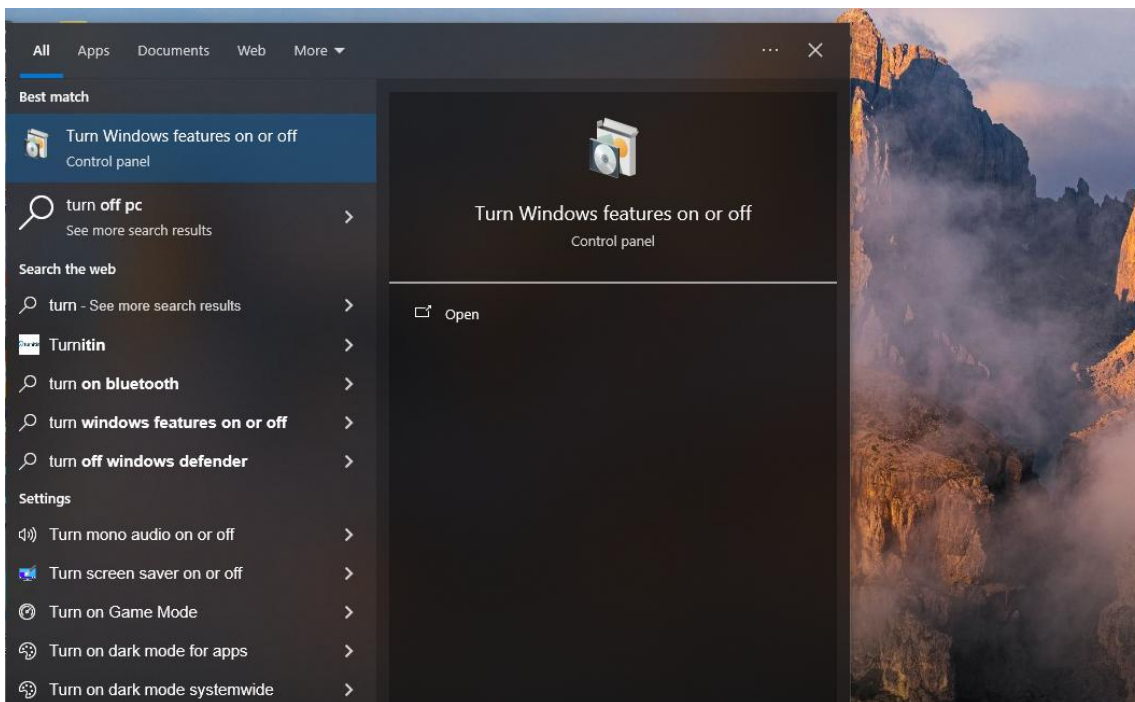
Configuration of Hyper-V on Windows of VMs

Date:**Aim:**

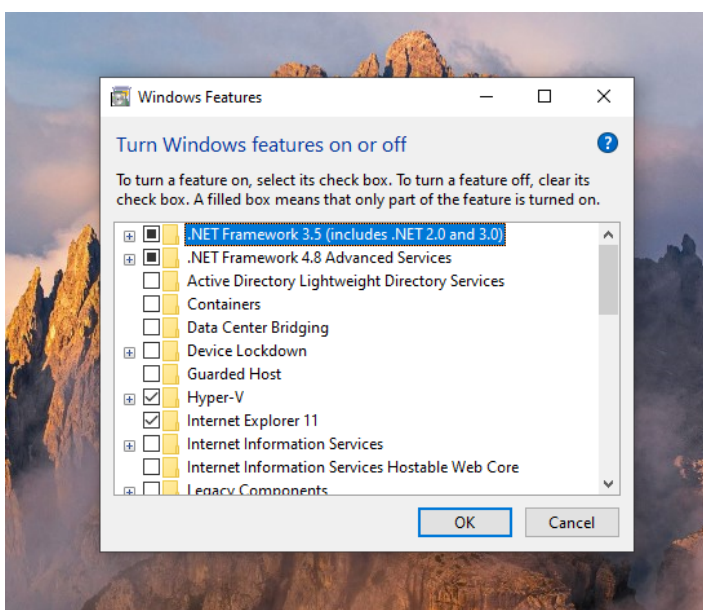
To configure Hyper-V on Windows for creating and managing virtual machines (VMs), enabling efficient virtualization and resource utilization.

Steps:

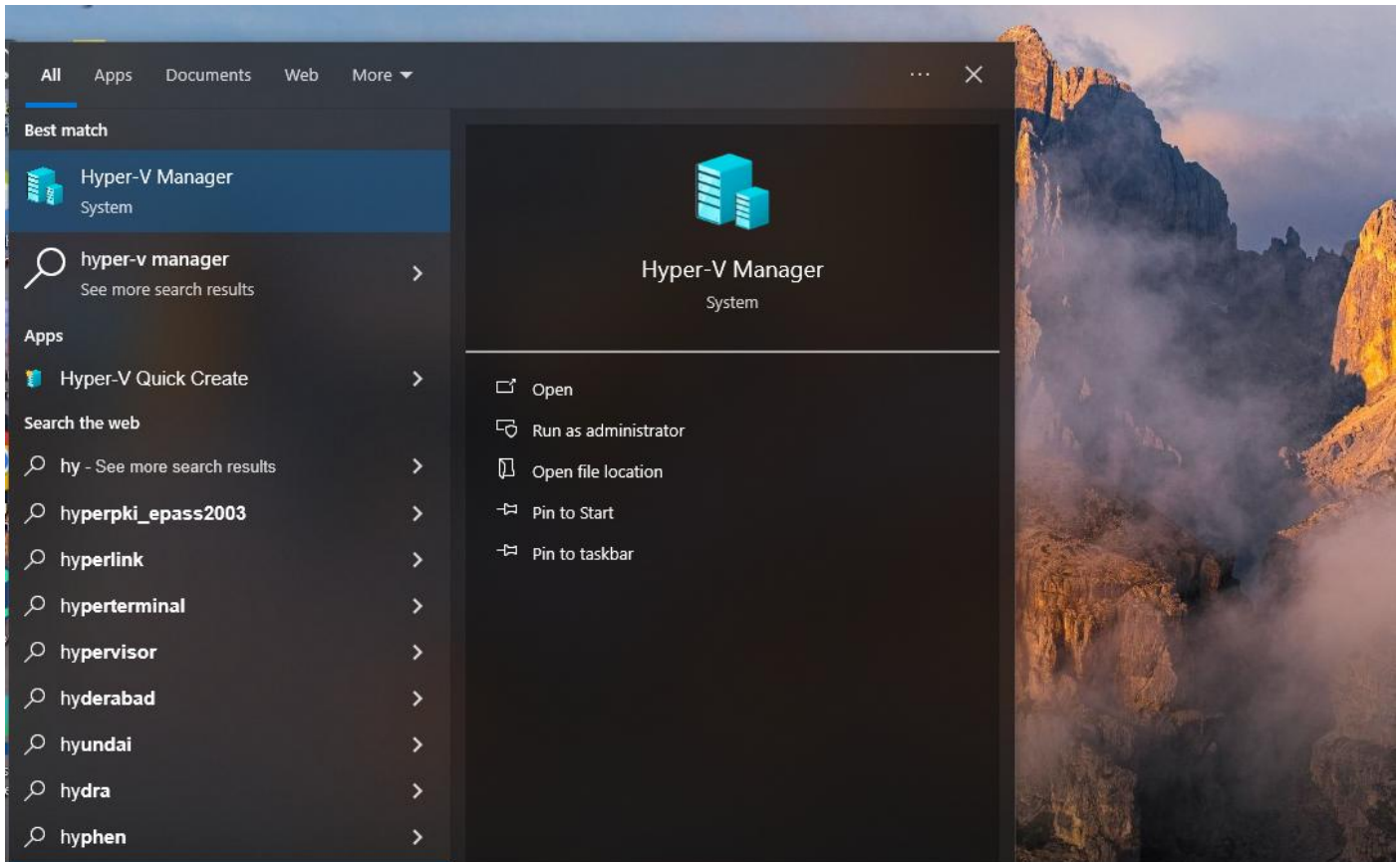
1) Click Turn Windows features on or off



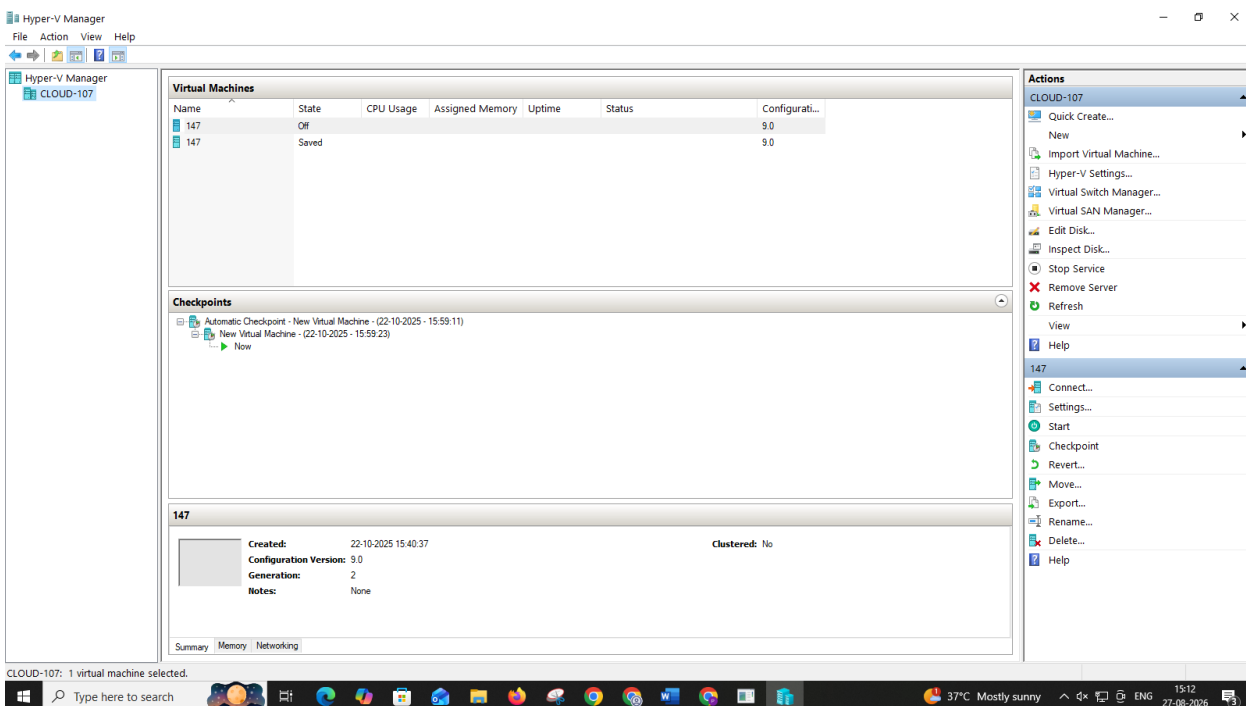
2) Find Hyper-V select all hyper -V and restart



3) After your PC restarts Press the Windows key and search: Hyper-V Manager



4) The hypervisor manager will open



RESULT:

Thus the configuration Hyper-V on Windows for creating and managing virtual machines (VMs), enabling efficient virtualization and resource utilization has been executed successfully.

Ex.No: 7b Creation of VM, Powershell commands, Management

Date:

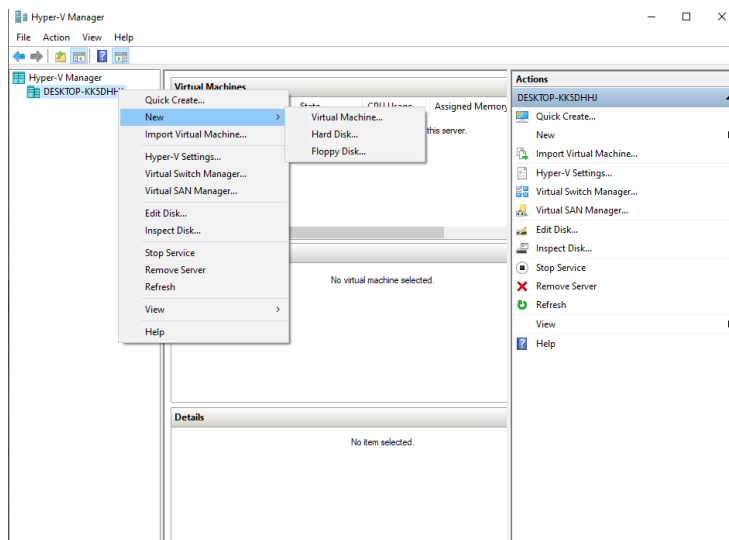
Aim:

To create and manage Virtual Machines (VMs) using Hyper-V, including PowerShell commands for automation and efficient administration.

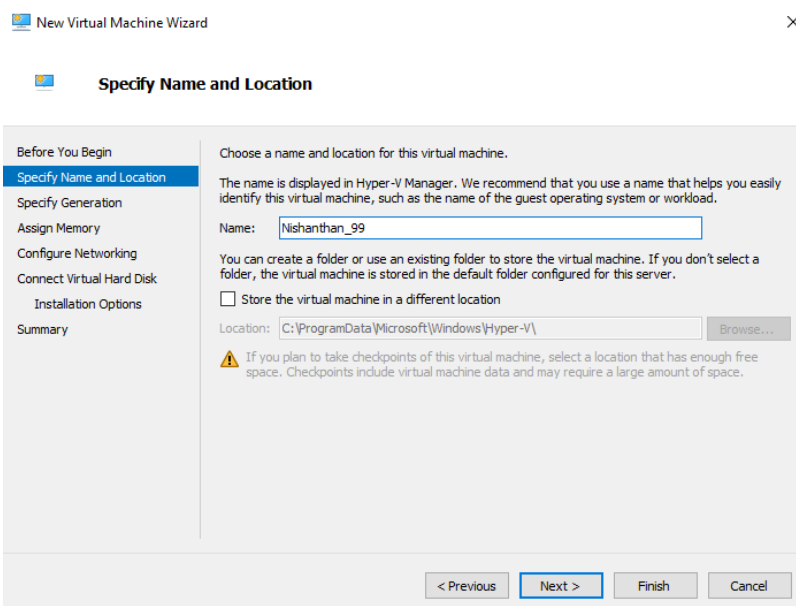
Procedure:

(Vm installation without powershell)

1)open Tools-> hyper v manager-> right click on server and click create new vm



2)Specify name and location click next.



3. Now Specify generation 2

New Virtual Machine Wizard

Specify Generation

Before You Begin
Specify Name and Location
Specify Generation
Assign Memory
Configure Networking
Connect Virtual Hard Disk
Installation Options
Summary

Choose the generation of this virtual machine.

☐ Generation 1
This virtual machine generation supports 32-bit and 64-bit guest operating systems and provides virtual hardware which has been available in all previous versions of Hyper-V.

☒ Generation 2
This virtual machine generation provides support for newer virtualization features, has UEFI-based firmware, and requires a supported 64-bit guest operating system.

⚠ Once a virtual machine has been created, you cannot change its generation.

[More about virtual machine generation support](#)

< Previous **Next >** Finish Cancel

3)Assign the memory for the vm(for safety give as 2gb)

New Virtual Machine Wizard

Assign Memory

Before You Begin
Specify Name and Location
Specify Generation
Assign Memory
Configure Networking
Connect Virtual Hard Disk
Installation Options
Summary

Specify the amount of memory to allocate to this virtual machine. You can specify an amount from 32 MB through 251658240 MB. To improve performance, specify more than the minimum amount recommended for the operating system.

Startup memory: MB

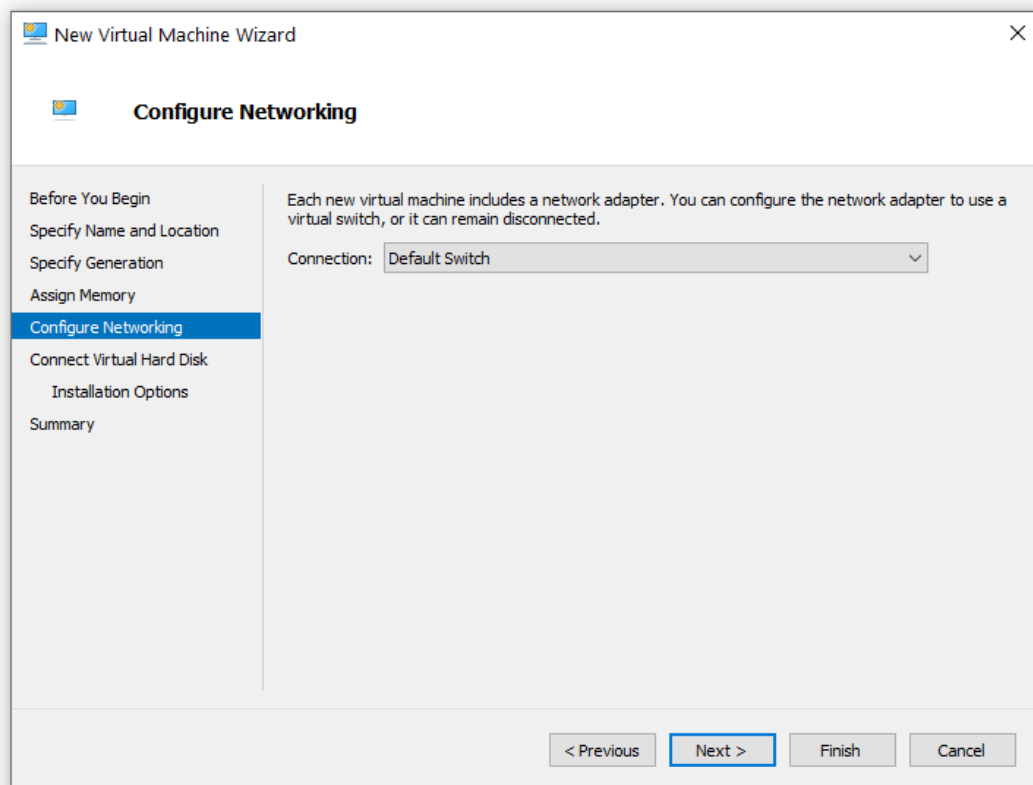
☒ Use Dynamic Memory for this virtual machine.

i When you decide how much memory to assign to a virtual machine, consider how you intend to use the virtual machine and the operating system that it will run.

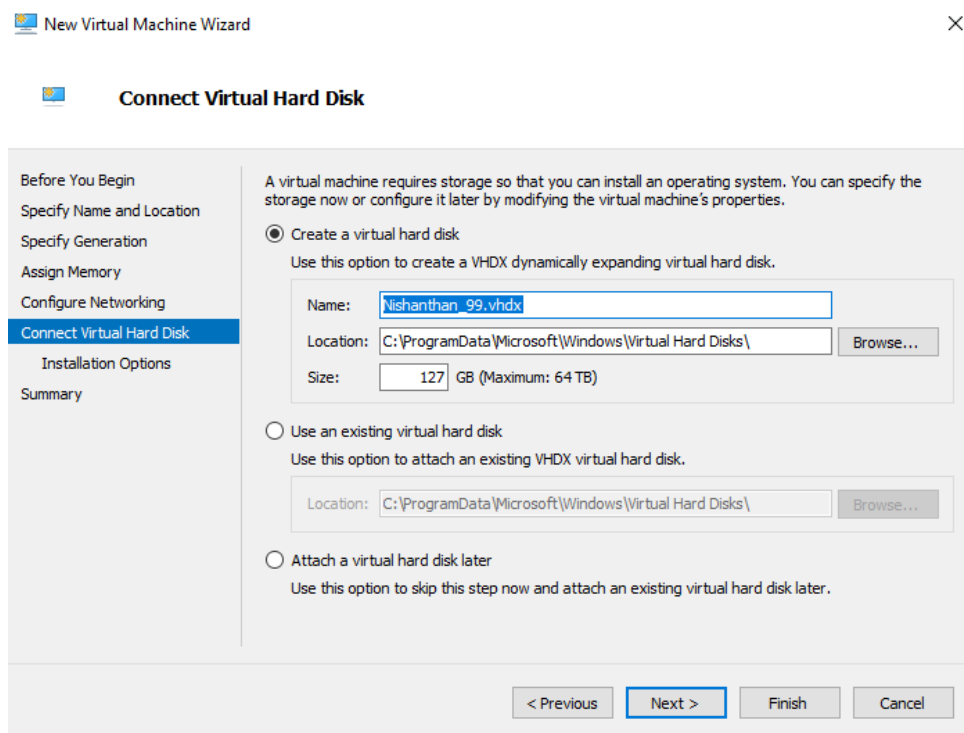
< Previous **Next >** Finish Cancel

4)Configure the network click on Default Switch

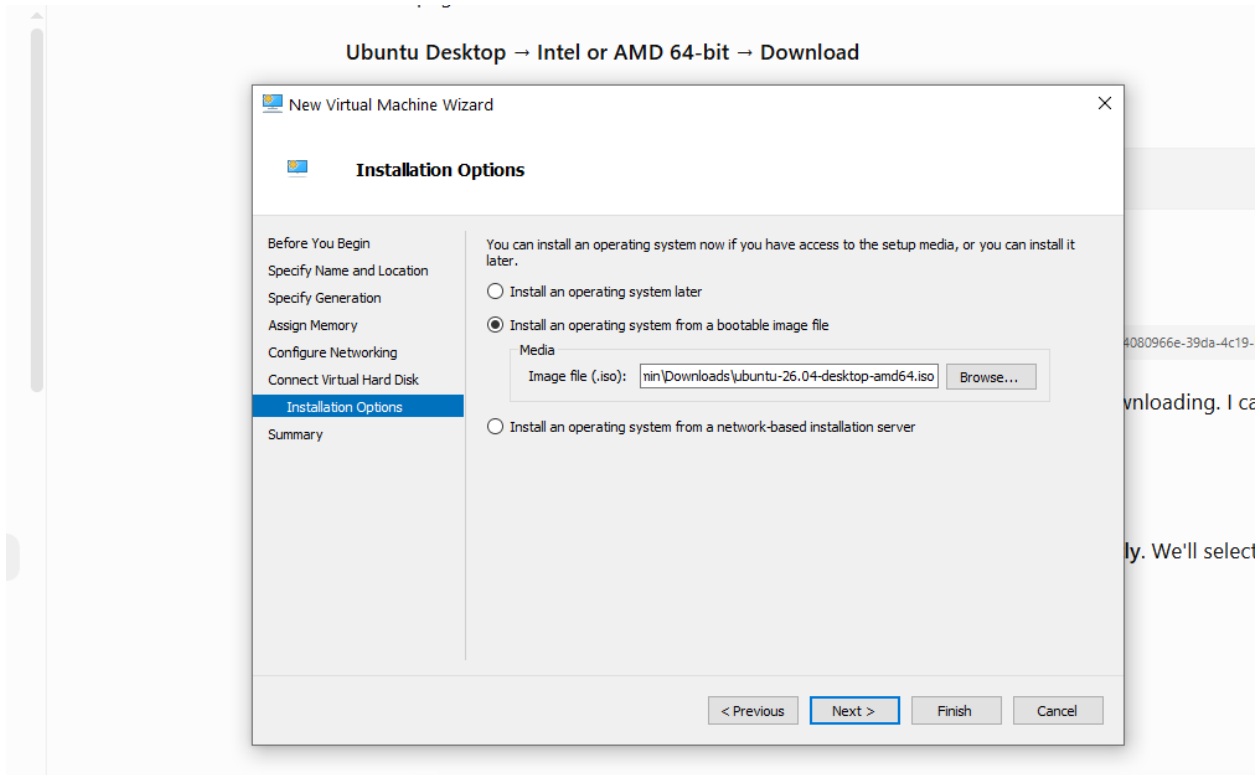
Then click:



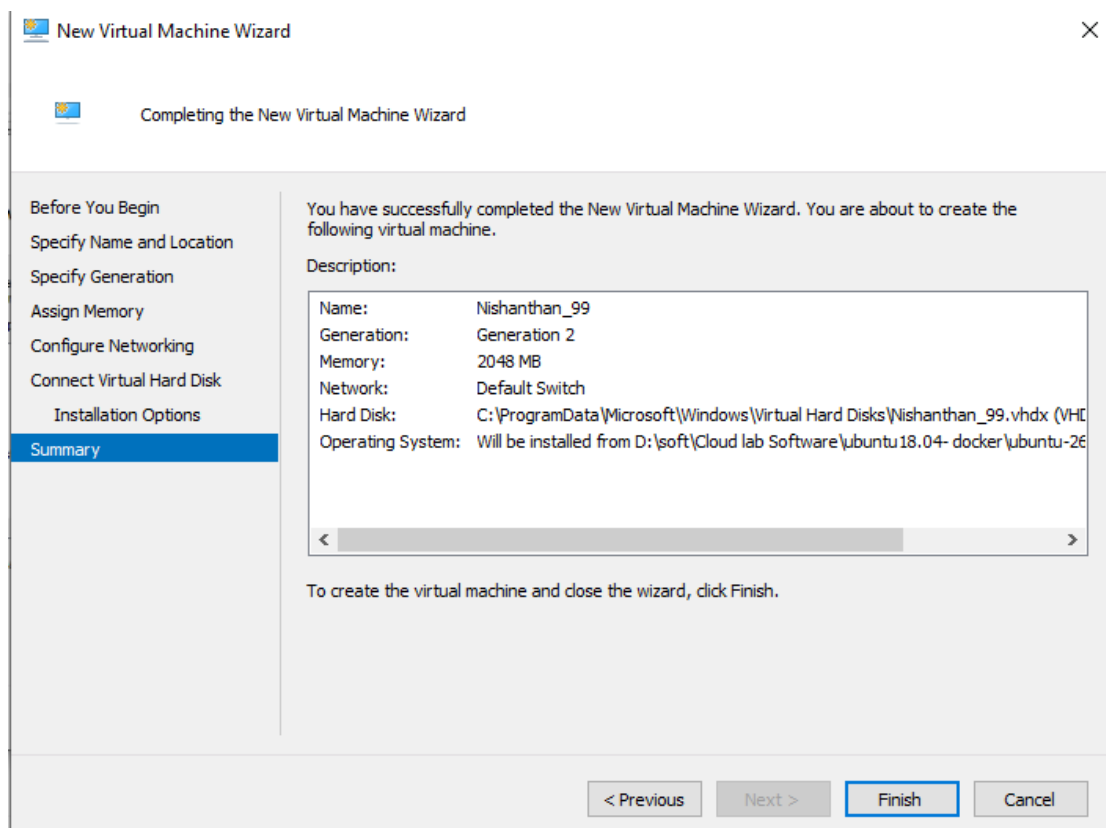
5)Connect virtual harddisk click next



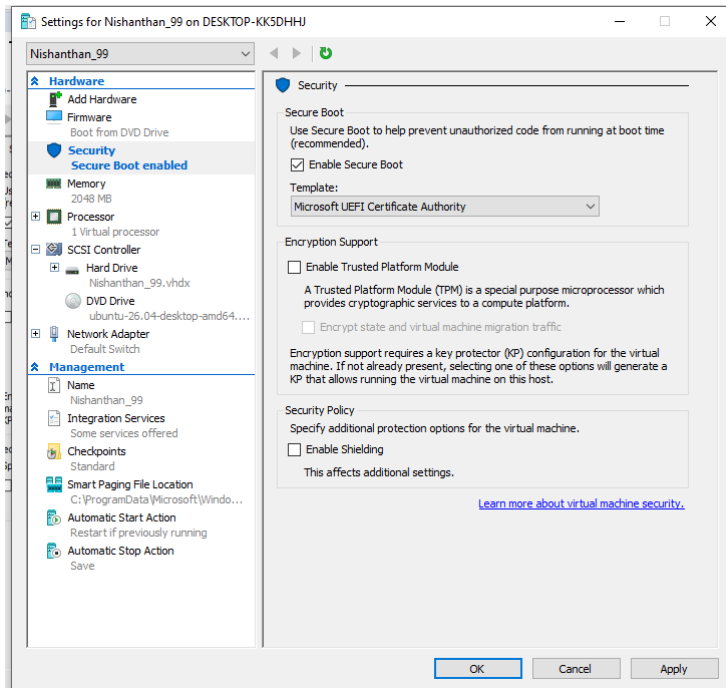
6) Import the iso file which os need to install on the vm(here ubuntu)



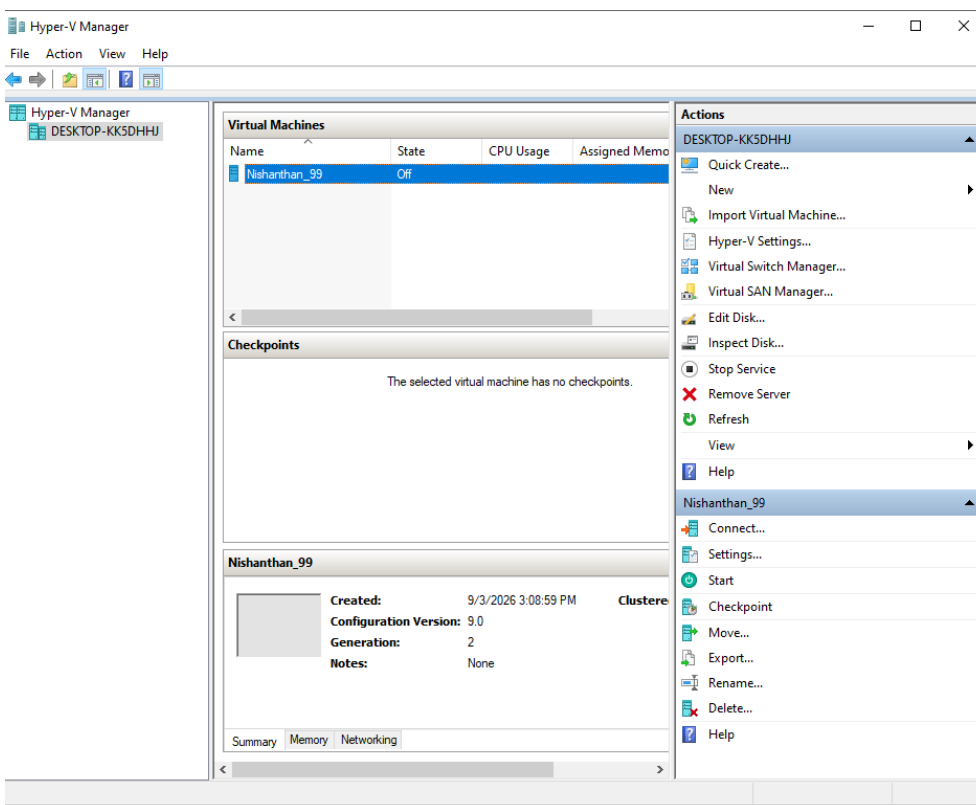
7) Click finish



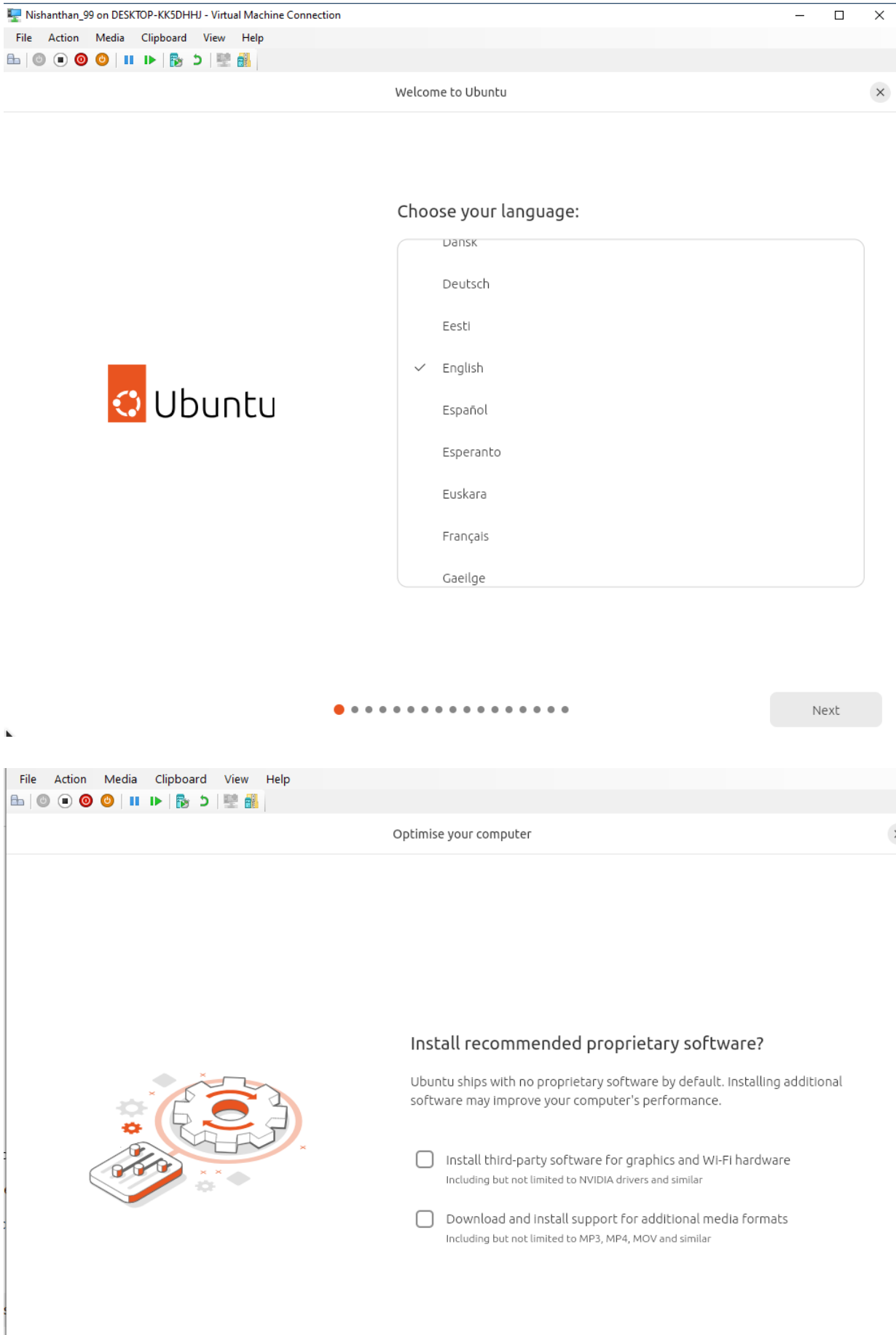
8) Change the Security setting - Change Template to MicrosoftUEFI



9) we can start the the vm in hyper-V manager by using connect option in the right panel after starting the VM



10) Once the VM starts it ask for setup and follow the steps in the screenshots :



After finishing all the steps it will start to install and it will ask to restart after restarting the ubuntu screen will appear

Vm installation using Powershell **Procedure :**

Get-VMSwitch

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> Get-VMSwitch

Name          SwitchType NetAdapterInterfaceDescription
-----
Default Switch Internal
```

1) Create vm with name and space in gb with virtual hard disk creation

```
PS C:\Windows\system32> New-Item -ItemType Directory -Path "C:\Hyper-V" -Force

Directory: C:\

Mode                LastWriteTime         Length Name
-----
d-----         27-08-2026         16:34         Hyper-V

PS C:\Windows\system32> New-VM -Name "Pradheeban_Ubuntu_PS" -MemoryStartupBytes 2GB -Generation 2 -NewVHDPath "C:\Hyper-V\Pradheeban_Ubuntu_PS.vhdx" -NewVHDSIZEBytes 30GB

Name          State CPUUsage(%) MemoryAssigned(M) Uptime      Status      Version
-----
Pradheeban_Ubuntu_PS Off      0           0           00:00:00 Operating normally 9.0
```

2)Assign virtual switch to the vm give the switch as we already got in the get switch command

```
PS C:\Windows\system32> Connect-VMNetworkAdapter -VMName "Pradheeban_Ubuntu_PS" -SwitchName "Default Switch"
PS C:\Windows\system32>
```

3)import the iso file which os need to install on vm

```
PS C:\Windows\system32> Add-VMdvdDrive -VMName "Pradheeban_Ubuntu_PS" -Path "C:\Users\Admin\Downloads\ubuntu-26.04-desktop-amd64.iso"
PS C:\Windows\system32>
```

4) Start the vm

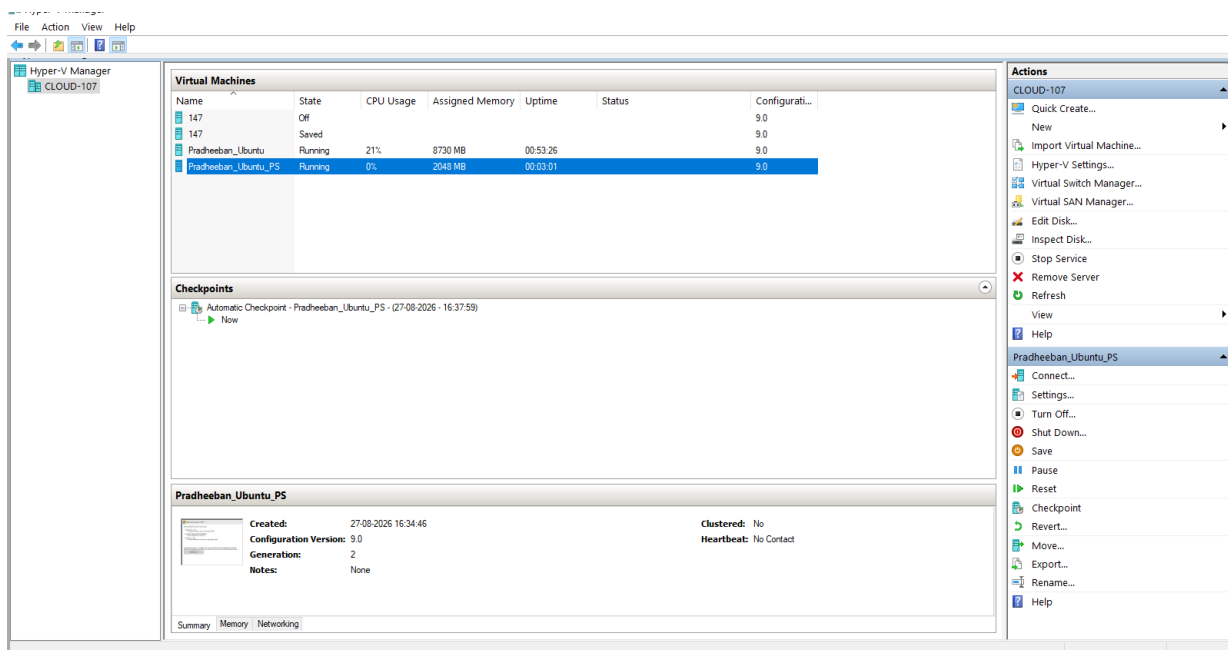
```
PS C:\Windows\system32> Set-VMFirmware -VMName "Pradheeban_Ubuntu_PS" -FirstBootDevice (Get-VMCdDrive -VMName "Pradheeban_Ubuntu_PS")
PS C:\Windows\system32> Start-VM -Name "Pradheeban_Ubuntu_PS"
```

We can check the status

```
PS C:\Windows\system32> Get-VM -Name "Pradheeban_Ubuntu_PS"
```

Name	State	CPUUsage(%)	MemoryAssigned(M)	Uptime	Status	Version
Pradheeban_Ubuntu_PS	Running	24	2048	00:00:58.3130000	Operating normally	9.0

5) If you want to start the VM go to Hyper visor manager you can see the created VM now you can connect to the VM



Powershell commands

1) To see the status of VM and Stop the VM

```
PS C:\Windows\system32> Get-VM -Name "Pradheeban_Ubuntu_PS"
```

Name	State	CPUUsage(%)	MemoryAssigned(M)	Uptime	Status	Version
Pradheeban_Ubuntu_PS	Running	0	2048	00:04:11.8540000	Operating normally	9.0

```
PS C:\Windows\system32> Stop-VM -Name "Pradheeban_Ubuntu_PS"
```

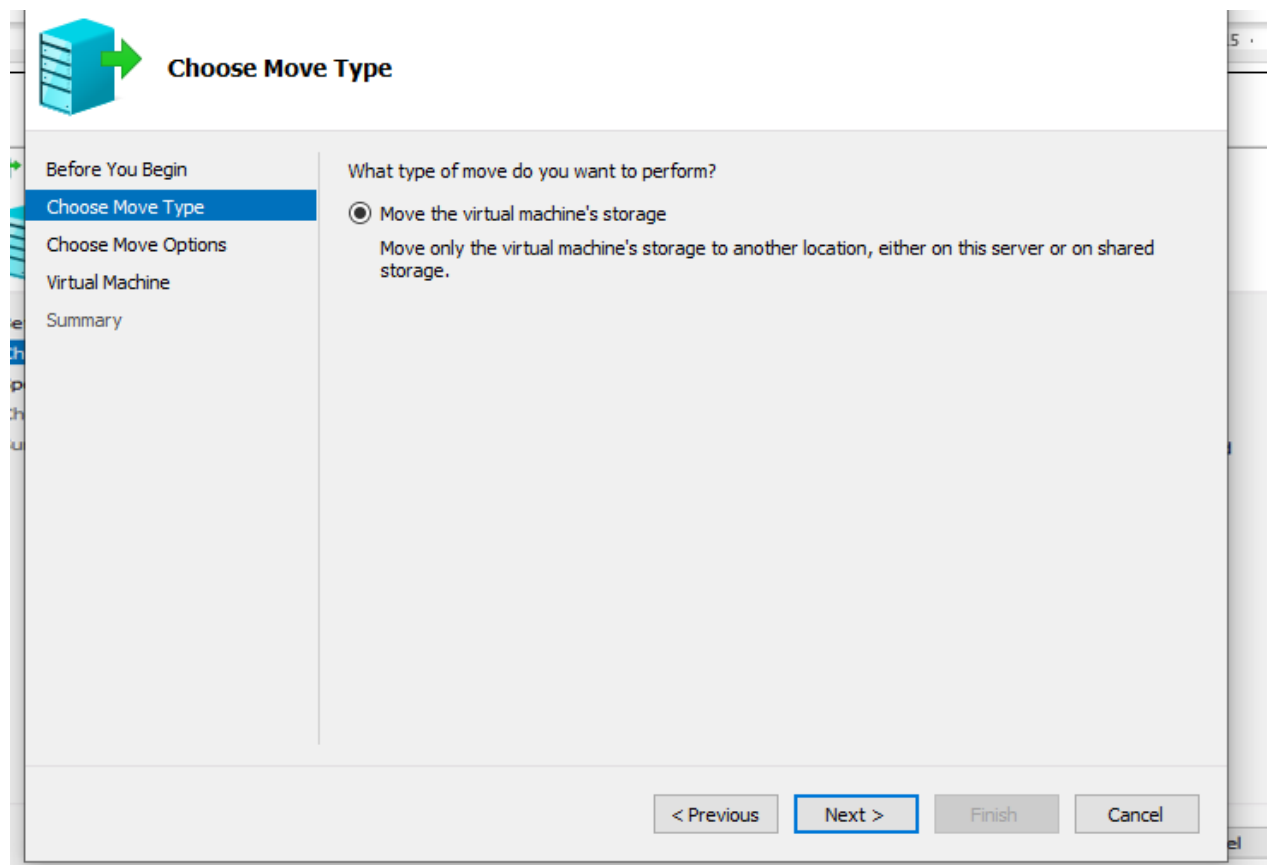
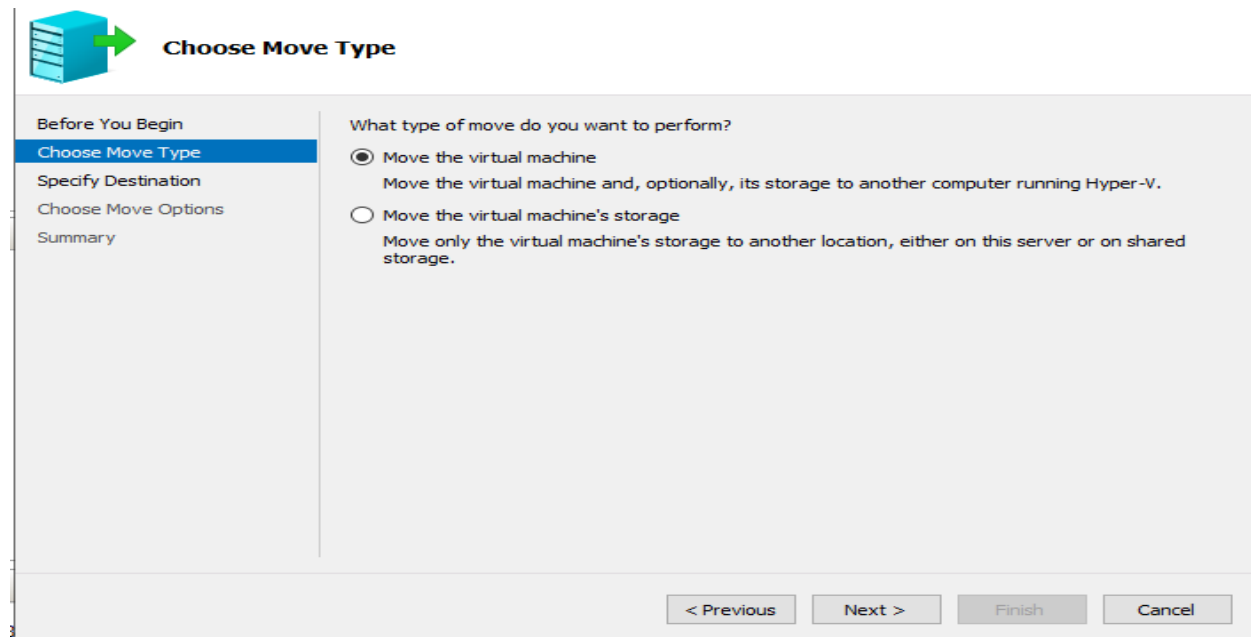
Confirm
Hyper-V can't shut down virtual machine "Pradheeban_Ubuntu_PS" because the Shutdown integration service is unavailable. You can turn off the virtual machine by se
physical machine. To avoid potential data loss, select [N]o, then pause or save the virtual machine. Do you want to turn off the virtual machine?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y"): y

```
PS C:\Windows\system32> Get-VM -Name "Pradheeban_Ubuntu_PS"
```

Name	State	CPUUsage(%)	MemoryAssigned(M)	Uptime	Status	Version
Pradheeban_Ubuntu_PS	Off	0	0	00:00:00	Operating normally	9.0

Migration :

Right click the vm and click move and follow the steps .





Choose Options for Moving Storage

Before You Begin

Choose Move Type

Choose Move Options

Virtual Machine

Summary

How do you want to move the storage?

- ☒ Move all of the virtual machine's data to a single location
This option allows you to specify one location to store all the virtual machine's items.
- ☐ Move the virtual machine's data to different locations
This option allows you to specify individual locations for each of the virtual machine's items.
- ☐ Move only the virtual machine's virtual hard disks
This option allows you to specify locations to move the virtual machine's virtual hard disks.

< Previous

Next >

Finish

Cancel

Move "Nishanthan_Ubuntu_PS" Wizard



Choose a new location for virtual machine

Before You Begin

Choose Move Type

Choose Move Options

Virtual Machine

Summary

Specify a location for the virtual machine's items.

New location

Folder: C:\Hyper-V\Migrated

Browse...

Available space: Not available

Current location

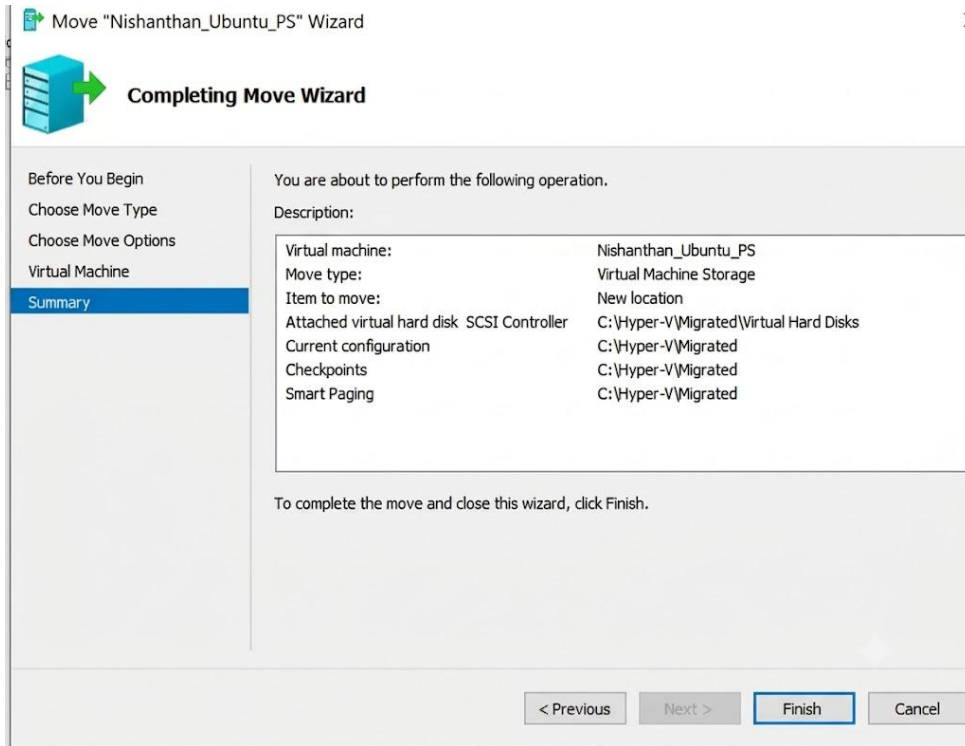
Size: 4 MB

< Previous

Next >

Finish

Cancel



Result:

Thus the creation and manage Virtual Machines (VMs) using Hyper-V, including PowerShell commands for automation and efficient administration has been executed successfully.